

STA363: Project

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EXECUTIVE SUMMARY AND CONCLUSION

This project analyzes NBA player salary data from the 2022–2023 season to understand how measurable basketball performance statistics relate to player compensation. The dataset includes (467) NBA players and a set of salary, demographic, box score, and advanced performance features. The response variable is player salary, and the main goal of this project is to predict salary using quantitative player characteristics.

The analysis begins with exploratory data analysis to understand the structure of the data and identify important relationships among variables. A correlogram was used to examine multicollinearity among the predictors, and several highly correlated features were removed or handled through later modeling approaches. The initial exploratory plots show that minutes played per game has a positive relationship with salary. However, the raw salary variable is highly skewed and shows increasing variability at higher salary levels, so a logarithmic transformation of salary was also considered to improve linearity and stabilize variance.

Several predictive models were then developed and compared. First, a K-Nearest Neighbors model was fitted using Leave-One-Out Cross-Validation to select the optimal value of K. The final KNN model used $K = 11$ and demonstrated moderate predictive performance, although it tended to underestimate salaries for some high paid players. Next, regression models were considered, including ordinary least squares regression, ridge regression, and lasso regression. Ridge regression helped reduce instability caused by multicollinearity, while lasso regression performed both shrinkage and feature selection. Among the regression models, lasso regression performed best, with a validation RMSE of (6.61) million dollars and only (13) features retained in the final model.

Finally, tree based models were used to improve interpretability and predictive performance. The single regression tree showed that minutes played per game (MP) was the first and most important splitting variable, followed by variables such as ‘age’, ‘points’, ‘three-point attempts’, ‘free throw rate’, and ‘BPM’. The tree had a cross-validated RMSE of (6.72) million dollars. Forest models were then fitted to improve stability and prediction accuracy. The bagged forest model achieved the lowest RMSE among the tree based models, with an out-of-bag RMSE of (5.81) million dollars, while the random forest model had an out-of-bag RMSE of (6.71) million dollars. Based on these results, the bagged forest model provided the strongest predictive performance overall.

Overall, the analysis suggests that NBA salary is strongly associated with playing time, scoring production, offensive responsibility, and overall player impact. Minutes played per game consistently appeared as one of the most important predictors across exploratory analysis, regression trees, and forest importance results. Although salary cannot be explained perfectly using only on court statistics, the models show that measurable performance variables provide meaningful predictive information. Among all models considered in this report, the bagged forest model appears to be the best predictive model, while the single regression tree provides the clearest interpretation of how player characteristics relate to salary.

AI STATEMENT

No generative AI tools were used in the process of conducting the analysis, performing the statistical modeling, or writing the substantive content of this report. All data analysis, coding, interpretation, and written explanations were completed by the equal contributions of Sihang Zhao & Haoxuan Sun, the joint authors of this work. The only digital tool used during the writing process was Grammarly, and it was used exclusively for minor grammar and spelling corrections.

Project Title: Predicting NBA Player Salaries Using Performance Statistics

INTRODUCTION

Professional basketball salaries are shaped by many factors, including player performance, team role, experience, market value, and contract structure. While some of these factors are difficult to measure directly, many aspects of player performance can be represented through numerical statistics. This project focuses on the question of how well measurable on-court performance statistics can be used to predict NBA player salary.

For this analysis, we use a dataset containing NBA player statistics and salaries from the 2022–2023 season. Each observation represents one NBA player, and the response variable is player salary measured in U.S. dollars. The dataset also includes variables such as age, minutes played per game, points, assists, rebounds, shooting statistics, usage rate, turnovers, and advanced metrics such as Box Plus/Minus. These variables are useful because they capture different dimensions of player contribution, including playing time, scoring production, efficiency, offensive involvement, and overall impact.

The main goal of this project is to build statistical and predictive models that estimate player salary based on available performance metrics. In addition to prediction, this project also aims to identify which variables are most strongly associated with salary. This is important because player salary in professional sports reflects how teams and the broader basketball market value different types of performance. By comparing multiple modeling approaches, we can evaluate which methods provide the best balance between accuracy, interpretability, and model complexity.

The report proceeds in several stages. First, exploratory data analysis is used to examine the structure of the dataset, identify multicollinearity, and understand the relationship between salary and key predictors. Next, a K-Nearest Neighbors model is developed for prediction. Then, regression models with and without selection are compared, including ordinary least squares regression, ridge regression, and lasso regression. Finally, tree based methods are applied, including a single regression tree, a bagged forest model, and a random forest model. These methods allow us to compare both interpretable and highly predictive approaches for modeling NBA player salary.

ANALYSIS PROCEDURES

The analysis began with data cleaning and preparation. The original dataset contained NBA player salaries, player identifiers, categorical variables, and quantitative performance statistics. Since the goal of this project was to predict salary using measurable basketball performance, player name, team, and position were not treated as primary explanatory variables in the initial cleaned dataset. The response variable was Salary, and the main predictors included numerical player statistics such as ‘Age’, ‘MP’, ‘AST’, ‘TRB’, ‘BPM’, and ‘TOV’. In later modeling sections, additional numerical and encoded variables were also used to allow the models to perform shrinkage, selection, or importance ranking.

After preparing the data, exploratory data analysis was conducted to better understand the relationships among variables. A raw data preview was first created to inspect the structure of the dataset. Then, a correlogram was used to check for multicollinearity among the predictors. This step was important because many basketball statistics are naturally correlated. For example, minutes played, field goal attempts, points, and usage rate are often related because players with larger roles usually play more minutes, take more shots, and score more points. Strong multicollinearity can make ordinary regression coefficients unstable and difficult to interpret.

Next, the relationship between Minutes Played per game and Salary was examined using scatter plots. The raw salary plot showed a positive relationship between minutes played and salary, but the pattern was not

perfectly linear and showed greater spread among players with higher salaries. Because of this, a logarithmic transformation of Salary was considered. The log-transformed salary plot showed a more linear relationship and more stable variance, suggesting that $\log(\text{Salary})$ may be more appropriate for some predictive modeling procedures.

The first predictive method used was K-Nearest Neighbors. KNN predicts a player's salary by finding players with similar performance characteristics and averaging the salaries of those nearby players. Because the dataset contained only (467) players, Leave-One-Out Cross-Validation was used to evaluate the model without losing too much training data. Different values of K from 2 to 50 were tested, and the value with the lowest validation RMSE was selected. Based on the validation results, $K = 11$ was chosen for the final KNN model.

The next stage of the analysis focused on regression models with and without selection. Ordinary least squares regression was first used as a baseline model. However, because the predictors showed strong multicollinearity, ridge regression and lasso regression were introduced. Ridge regression reduces coefficient instability by shrinking coefficients toward zero, while lasso regression can shrink some coefficients exactly to zero and therefore performs variable selection. For both ridge and lasso regression, validation RMSE was used to select the optimal tuning parameter λ . The final models were then compared based on RMSE, whether they performed feature selection, and the number of features retained.

The final modeling stage used tree based methods. A single regression tree was first constructed to provide an interpretable visual model for predicting salary. The tree split players into groups based on predictor values and assigned each group an average predicted salary. Cross-validated RMSE was used to evaluate the tree's predictive performance. Then, two forest models were fitted: a bagged forest model and a random forest model. Both models used (500) trees. The bagged forest model used all predictors at each split, while the random forest model used a random subset of predictors at each split. Out-of-bag RMSE was used to compare the forest models, and a variable importance plot was created to identify which predictors contributed most strongly to salary prediction.

Across all stages of the analysis, RMSE was used as the main performance metric. Because salary was often scaled by dividing by (1,000,000), RMSE values were interpreted in millions of U.S. dollars. Smaller RMSE values indicated better predictive accuracy. The final comparison focused on both predictive performance and interpretability, allowing us to identify which model best predicted salary and which variables were most important in explaining player compensation.

RESULTS ARE SHOWN BELOW

Section 1: Data and Goal Overview

For this analysis, we use a dataset containing NBA player statistics and salaries from the 2022–2023 season. The dataset consists of (467) observations and (52) variables, where each observation represents an individual NBA player’s statistics. The response variable is player salary, measured in U.S. dollars. While the dataset includes categorical variables such as player name, team, and position, these variables are excluded from the modeling process. Only quantitative performance statistics are used as predictors.

The goal of this project is to predict a player’s salary based solely on measurable on-court performance metrics. By focusing on numerical variables, we aim to assess the extent to which statistical productivity explains differences in player compensation.

The primary features considered include age, games played (GP), games started (GS), minutes per game (MP), field goals (FG), field goal percentage (FG%), three-point statistics, rebounds, assists, steals, blocks, turnovers, points per game, and other continuous box score or advanced metrics available in the dataset. These variables capture scoring efficiency, playing time, and overall player contribution.

This project is particularly interesting because salary determination in professional sports reflects performance valuation within a competitive market. By modeling salary as a function of quantitative performance indicators, we can evaluate how strongly observable productivity translates into financial compensation.

Section 2: EDA

Raw Data Glimpse

Table 1: Raw data preview (Part A)

Player.Name	Salary	Position	Age	Team	MP	FGA	X3PA	X2P.	FT.
Stephen Curry	48070014	PG	34	GSW	34.7	20.2	11.4	0.58	0.92
John Wall	47345760	PG	32	LAC	22.2	9.9	3.2	0.46	0.68
Russell Westbrook	47080179	PG	34	LAL/LAC	29.1	13.6	3.9	0.49	0.66
LeBron James	44474988	PF	38	LAL	35.5	22.2	6.9	0.58	0.77
Kevin Durant	44119845	PF	34	BRK/PHO	35.6	18.3	4.9	0.62	0.92
Bradley Beal	43279250	SG	29	WAS	33.5	17.6	4.4	0.55	0.84

Table 2: Raw data preview (Continued)

Player.Name	Salary	TRB	PF	PTS	TS.	X3PAr	FTTr	TRB.	AST.	STL.	BLK.
Stephen Curry	48070014	6.1	2.1	29.4	0.66	0.56	0.25	9.7	30.0	1.3	0.9
John Wall	47345760	2.7	1.7	11.4	0.50	0.32	0.33	6.8	35.3	1.8	1.4
Russell Westbrook	47080179	5.8	2.2	15.9	0.51	0.29	0.32	10.8	38.6	1.7	1.3
LeBron James	44474988	8.3	1.6	28.9	0.58	0.31	0.27	12.5	33.5	1.2	1.4
Kevin Durant	44119845	6.7	2.1	29.1	0.68	0.27	0.39	10.5	24.5	1.0	3.4
Bradley Beal	43279250	3.9	2.1	23.2	0.59	0.25	0.26	6.5	26.6	1.3	1.7

As shown in Table 1 & 2 above, our raw data includes in total of (22) columns, featuring (1) response variable (Y: Salary in US Dollar), and (21) features (3 categorical and 18 numerical). However, there are some issues within our feature set, particularly potential multicollinearity among the predictors. Therefore, we are going to conduct a multicollinearity check across all features and remove those with strong correlations with each other.

show strong correlations with each other in order to reduce redundancy and improve model stability.

The Data

Table 3: The First 10 Rows of Our Data

Salary	Age	MP	AST.	TRB	BPM	TOV.
48070014	34	34.7	30.0	6.1	7.5	12.5
47345760	32	22.2	35.3	2.7	-1.2	17.1
47080179	34	29.1	38.6	5.8	0.2	18.4
44474988	38	35.5	33.5	8.3	6.1	11.6
44119845	34	35.6	24.5	6.7	7.1	13.4
43279250	29	33.5	26.6	3.9	1.8	12.9
42492492	31	33.6	19.6	6.5	6.1	8.1
42492492	32	34.6	24.2	6.1	2.8	13.4
42492492	28	32.1	33.2	11.8	8.5	13.2
42492492	32	36.3	35.0	4.8	7.1	11.7

After data processing, we selected (6) features that most strongly affect our prediction while showing no strong correlation with each other, and created the final dataset used in this report. As shown above in Table 3, we present the first (10) rows of our cleaned dataset.

To help our client better understand the acronyms used in the dataset, Table 4 provides the full names and definitions of each variable included in the model.

For the complete dataset used in this analysis, please refer to Appendix A at the end of this report.

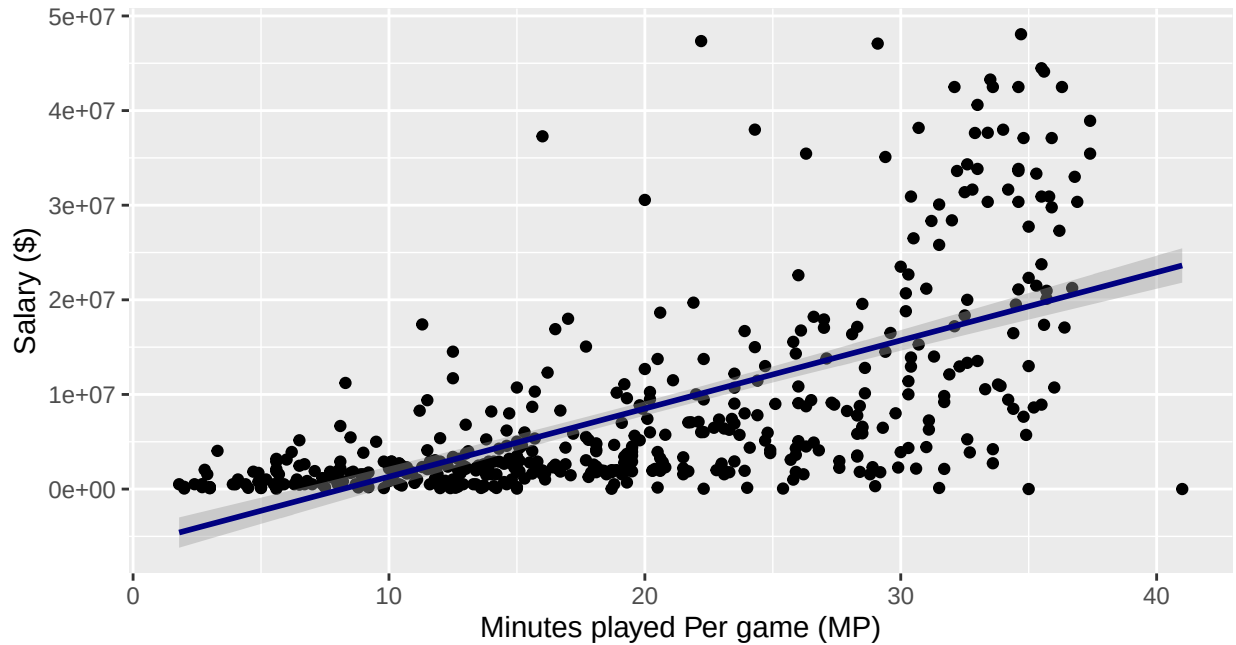
Table 4: Variable Definitions

Acronym	Explanation
Salary	Annual player salary (USD)
Age	Player age (years)
MP	Minutes played per game
AST	Assists per game
TRB	Total rebounds per game
BPM	Box Plus/Minus — overall impact metric
TOV	Turnovers per game

Data Exploration

Research Question: How does the players' salary change over MP (Minutes played Per game)?

Figure 1

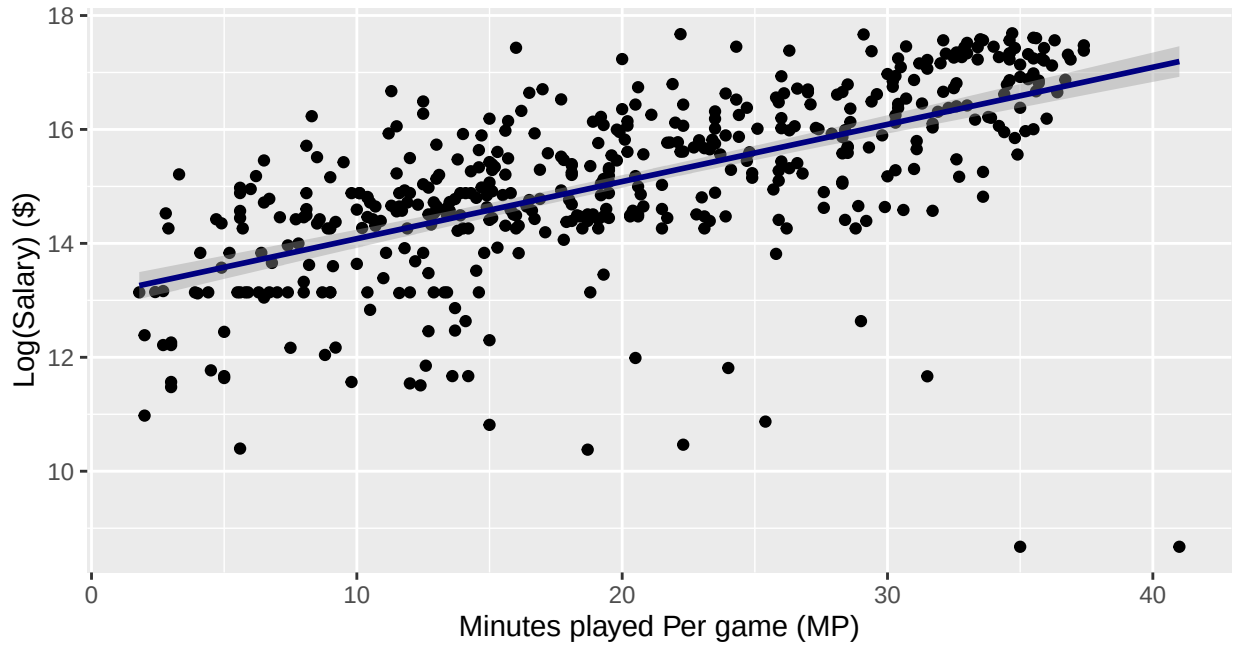


The relationship between minutes played per game and salary

As shown in Figure 1 above, although there is a clear positive relationship between Minutes Played per game and Salary, the pattern does not appear strictly linear. The spread of the data increases as Minutes Played increases, indicating potential heteroskedasticity, and the relationship appears slightly curved rather than perfectly linear. This suggests that a linear model on the raw Salary variable may not be the most appropriate specification.

Therefore, it seems that applying a logarithmic transformation to Salary may provide a better fit. By modeling $\log(\text{Salary})$ instead of Salary, we can stabilize the variance, reduce skewness in the response variable, and potentially achieve a more linear relationship between Minutes Played and Salary. Consequently, we will repeat the same analysis using $\log(\text{Salary})$ as the response variable.

Figure 2



The relationship between minutes played per game and log salary

The log transformation appears to substantially improve the linearity of the relationship between Minutes Played per game and Salary. As shown in Figure 2, when modeling $\log(\text{Salary})$ against Minutes Played, the data points align more closely around the fitted regression line compared to the raw Salary model in Figure 1.

In addition, the variance of the residuals appears more stable across different levels of Minutes Played. The relationship now shows a clearer linear trend, suggesting that a linear regression model with $\log(\text{Salary})$ as the response variable is more appropriate than using raw Salary.

Based on the scatterplot and the fitted regression line, players' salary increases as Minutes Played per game increases. There is a clear positive relationship between playing time and compensation: players who average more minutes per game tend to earn higher salaries.

When examining the raw Salary model in Figure 1, the relationship appears upward sloping but exhibits increasing variability at higher levels of Minutes Played, indicating potential nonlinearity. After applying a logarithmic transformation to Salary, the relationship becomes more linear and the variance appears more stable. The fitted regression line in the $\log(\text{Salary})$ model confirms a strong positive association between Minutes Played and salary.

Interpreting the log-scale model, this suggests that each additional minute played per game is associated with a proportional (percentage) increase in salary rather than a constant dollar increase.

Section 3: Prediction

In this section, we describe our predictive methodology, the model employed, and the validation techniques used to evaluate its performance. We utilized K-Nearest Neighbors (KNN) algorithm to build our model. KNN is a distance-based, non-parametric predictive algorithm.

Section 3.1: KNN Procedures

To predict the salary of a given player based on their characteristics, KNN proceeds as follows:

First, the algorithm computes distance measures to determine how similar the player in the test data is compared to all players in the training dataset. This is done using Gower's distance.

Then, for a chosen value of K , the 'K' number of players in the training data whose feature profiles are the closest to the player in the test dataset.

Finally, the predicted salary is computed as the average salary of these K neighboring players.

In other words, the model assumes that players with similar performance characteristics tend to earn similar salaries, and it uses the observed salaries in the training data to generate the prediction.

Section 3.2: Motivation

The primary motivation for choosing KNN is its flexibility. Salary structure in the NBA is unlikely to follow a strictly linear relationship with performance metrics. In other words, equal changes in performance statistics may not translate into equivalent changes in salary across different levels of the feature space.

Moreover, salary structures may differ substantially between role players and star players. By making predictions based on the player's nearest neighbors, KNN effectively allows predictions to be made within local groups and tiers of players. This local averaging approach is intuitive and aligns well with how player valuation may operate in practice.

To optimize the performance of our KNN model, we must carefully determine the value of K . To determine the most appropriate value for K , we implemented a cross-validation technique: Leave-One-Out Cross-Validation (LOOCV).

We chose this re-sampling method since our dataset's size is highly constrained, containing information for only (467) players. With such a limited sample size, dividing into both train and validation data would reduce the data available for training, making it an impractical approach.

Section 3.3: LOOCV

In LOOCV, the model iteratively selects a single player as the validation set while training on the remaining observations. In this report, the training data for our KNN using LOOCV will always be (466) rows. This process is repeated for every player in the dataset. After we finish all the iterations, we retrieve our predictions on each row of data. To measure prediction accuracy, we calculate Root Mean Squared Error (RMSE) by averaging the prediction errors across all iterations.

To tune our K , we applied this LOOCV process for every potential K value ranging from 2 to 50, a conventional range of choice for K . We then select the K that minimizes the overall RMSE. Based on Figure 3 below, we mapped the calculated RMSE results against the choices of K and come to a conclusion that we should choose $K = 11$.

Figure 3

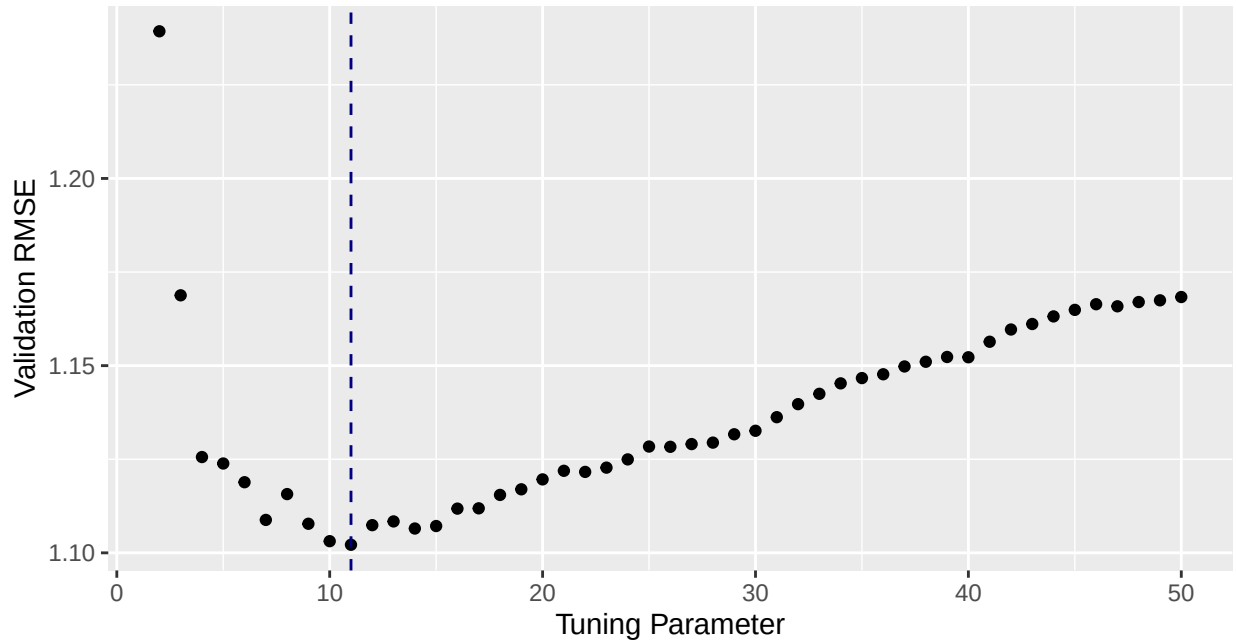


Figure 4

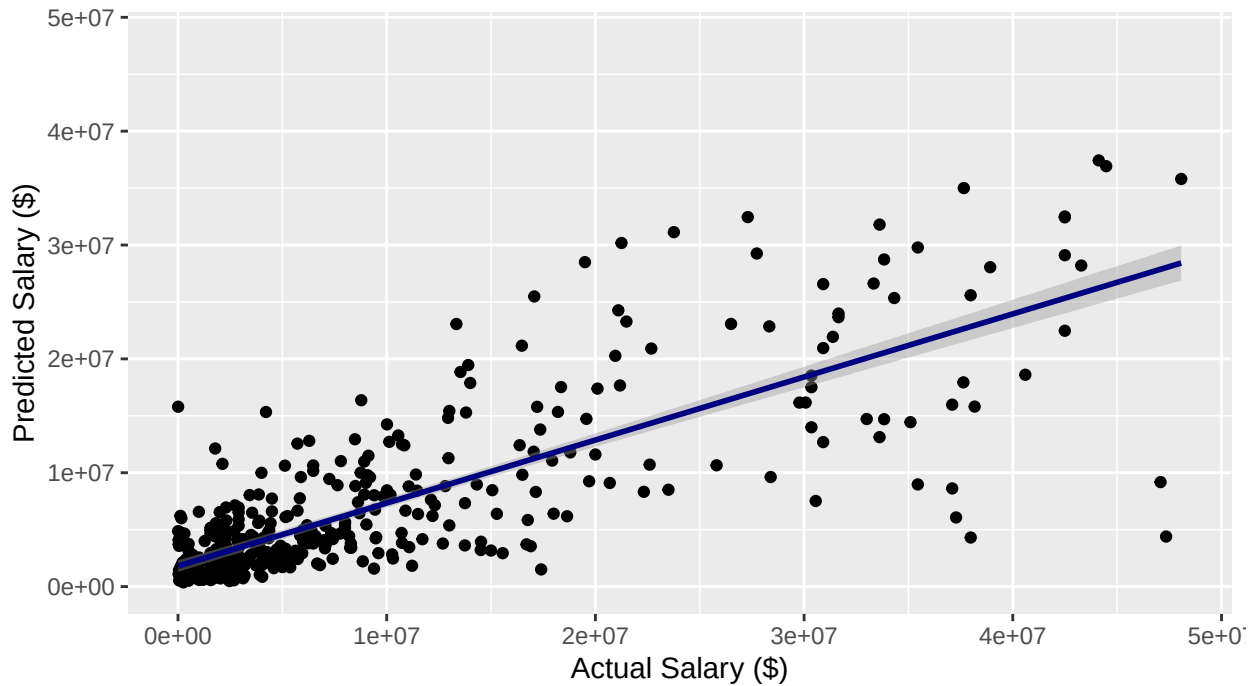


Figure 4 above is a graph that plots the actual player salaries against the predicted salaries generated by our KNN prediction algorithm. Using the log transformation in the prediction helps reduce the skewness present in the salary data and allows for a more meaningful comparison between the prediction and actual data, we then exponentiated the prediction results to map the actual salary data. Overall, the model demonstrates moderate predictive performance, with relatively high variance for players with higher wages.

However, one notable pattern is the presence of frequent underestimations, where the predicted salary is

lower than the actual salary. This suggests that the model may be biased toward predicting larger values, particularly for certain players. Improving the model could involve further tuning the value of K , incorporating additional relevant features, or experimenting with alternative modeling approaches to reduce these systematic prediction errors.

Section 4: Regression with/without Selection

In this section, instead of hand-picking features based on the correlogram, we bring all features back into the analysis and let the regression model handle the selection and shrinkage process for us. From the earlier EDA, we found that many predictors are strongly correlated with one another, which may cause multicollinearity and lead to unstable coefficient estimates in an ordinary linear regression model. To cross out this issue, we introduce ridge regression and lasso regression, two penalized regression methods that add a penalty term to the loss function. Ridge regression helps reduce the effect of multicollinearity by shrinking coefficients, while lasso regression can further simplify the model by shrinking some coefficients all the way to zero. Therefore, in this section, we compare regression models with and without feature selection in order to examine how these methods affect model fit and prediction of NBA player salary.

Section 4.1: Regression without Selection

For comparison, we first fit The Least Squares Line Regression model (LSLR) using all numerical predictors and no feature selection. This provides a useful baseline before introducing ridge regression. However, as shown earlier in the correlogram, many of the predictors are strongly correlated with one another, which leads to multicollinearity in the linear regression. When multicollinearity is present, LSLR may produce unstable coefficient estimates, unrealistically large standard errors, and coefficients that are difficult to interpret individually, as shown in Table 5 below. More importantly, these issues can reduce the model's predictive reliability. For this reason, LSLR is not ideal for our data. To address this problem, we introduce ridge regression, which keeps all predictors in the model but changes the way the coefficients are estimated by adding a penalty term. This shrinkage helps reduce the effect of multicollinearity and produces more stable coefficient estimates, making ridge regression more appropriate than ordinary least squares regression for this dataset.

*Table 5 presents the first 10 coefficient estimates from the full ordinary linear regression model. The complete coefficient table is provided in the Appendix C at the end of this report.

Table 5: First 10 coefficient estimates from the full ordinary linear regression model

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3937945.5	8977907.05	-0.4386	0.6612
PositionPF	-1035172.4	1363116.21	-0.7594	0.4481
PositionPG	1890715.9	2089278.61	0.9050	0.3661
PositionPG-SG	4439934.9	6290111.50	0.7059	0.4808
PositionSF	1342520.5	1588642.79	0.8451	0.3987
PositionSF-PF	736689.5	6883075.00	0.1070	0.9148
PositionSF-SG	-195446.3	8175409.28	-0.0239	0.9809
PositionSG	-186726.7	1750160.18	-0.1067	0.9151
PositionSG-PG	221003.9	8057332.72	0.0274	0.9781
Age	798813.1	90883.71	8.7894	0.0000

Ridge Regression

Since the LSLR model is not ideal for our data, we now turn to ridge regression as an alternative. Ridge regression does not change the form of the linear regression model, but it does change how the coefficients are estimated. Instead of minimizing only the residual sum of squares,

$$RSS = (Y - X_D \hat{\beta})^T (Y - X_D \hat{\beta}),$$

ridge regression minimizes

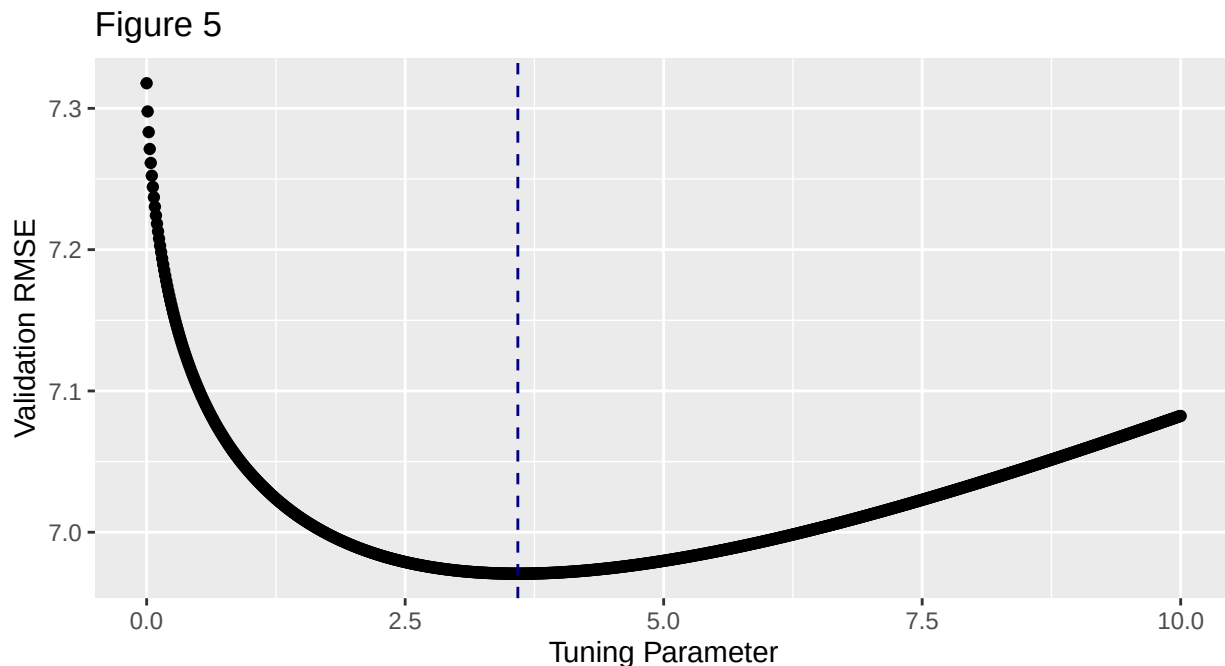
$$(Y - X_D \hat{\beta})^T (Y - X_D \hat{\beta}) + \lambda \hat{\beta}^T \hat{\beta},$$

where $\lambda \geq 0$ is a tuning parameter. Equivalently, the penalty term may be written as

$$\lambda \sum_{j=1}^p \hat{\beta}_j^2.$$

This additional penalty shrinks the coefficient estimates toward zero, which helps reduce the instability caused by multicollinearity. Therefore, while ridge regression may not fit the training data as closely as the LSLR model, it often produces more stable coefficient estimates and better predictive performance when predictors are highly correlated. The amount of shrinkage is controlled by λ : when $\lambda = 0$, ridge regression reduces to the LSLR solution, and as λ increases, the coefficient estimates are shrunk more strongly toward zero. We choose the optimal value of λ by comparing validation RMSE across a range of possible values, as shown in Figure 5 below.

Note that Salary was divided by 1,000,000 before fitting the ridge regression model, so all RMSE values and coefficient estimates are reported on the scale of millions of U.S. dollars.



Validation RMSE values for different tuning parameters.
Smallest RMSE at lambda = 3.59 . RMSE is reported in millions of USD.

Table 6: Tuning results for the ridge regression model

Metric	Value
Optimal lambda	3.5900
Minimum validation RMSE	6.9707

Figure 5 & Table 6 above shows the validation RMSE for different values of the tuning parameter λ . The validation RMSE decreases at first as λ increases from 0, which suggests that a moderate amount of shrinkage improves predictive performance by reducing the instability caused by multicollinearity. However, after a certain point, the validation RMSE begins to increase, indicating that too much shrinkage hurts model performance. The minimum validation RMSE occurs at $\lambda = 3.59$, so we select this value as the optimal tuning parameter for the final ridge regression model. Since the response variable was divided by 1,000,000 before fitting the model for better prediction purpose, the reported validation RMSE is measured in millions of U.S. dollars rather than raw dollars.

Using the optimal tuning parameter $\lambda = 3.59$, we fit the final ridge regression model. Because Salary was scaled by dividing by 1,000,000, the ridge regression coefficients should be interpreted with respect to salary measured in millions of dollars. Table 7 presents the first 10 coefficient estimates from the fitted ridge model, while the complete coefficient table is provided in the Appendix.

*Table 7 below shows the first 10 coefficient estimates from the fitted ridge model, while the complete coefficient table is provided in the Appendix D at the end of this report.

Table 7: First 10 coefficient estimates from the ridge regression model (Salary measured in millions of U.S. dollars)

Variable	Coefficient
(Intercept)	-14.8107
PositionPF	-0.4325
PositionPG	0.7521
PositionPG-SG	3.8449
PositionSF	0.7571
PositionSF-PF	0.0520
PositionSF-SG	0.0238
PositionSG	-0.7370
PositionSG-PG	0.0552
Age	0.5700

Section 4.2: Regression with Selection

Although ridge regression improves stability by shrinking coefficients, it still keeps all predictors in the model. Therefore, we next consider lasso regression, which extends this idea by combining shrinkage with variable selection.

Like ridge regression, lasso adds a penalty term to the residual sum of squares. However, instead of using the squared coefficient penalty, lasso uses the sum of the absolute values of the coefficients. Specifically, lasso estimates the coefficient vector by minimizing

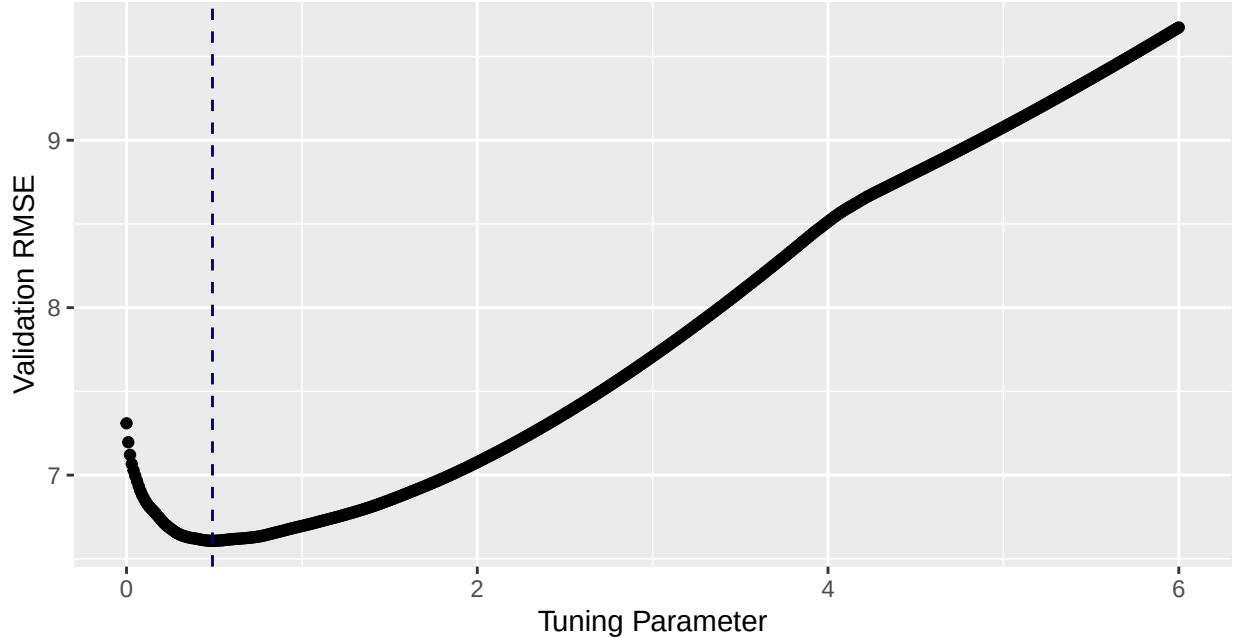
$$(Y - X_D \hat{\beta})^T (Y - X_D \hat{\beta}) + \lambda \sum_{j=1}^p |\hat{\beta}_j|,$$

where $\lambda \geq 0$ is a tuning parameter. Equivalently, this penalty can be written using the ℓ_1 norm as

$$RSS + \lambda \|\hat{\beta}\|_1.$$

The use of the ℓ_1 penalty gives lasso an important advantage over ridge regression: it can shrink some coefficient estimates all the way to zero. As a result, lasso performs both shrinkage and selection, producing a simpler and more interpretable model. Therefore, lasso regression is especially useful when we believe that only a subset of the predictors contributes meaningfully to the response. Since Salary was divided by 1,000,000 before fitting the model, all coefficient estimates and validation RMSE values in this subsection are reported in millions of U.S. dollars.

Figure 6



Validation RMSE values for Different Tuning Parameters. Smallest RMSE at lambda = 0.49

Table 8: Tuning results for the lasso regression model

Metric	Value
Optimal lambda	0.4900
Minimum validation RMSE	6.6072

Figure 6 & Table 8 above show the validation RMSE for different values of the lasso tuning parameter λ . The validation RMSE decreases at first as λ increases from 0, indicating that a moderate amount of penalization improves predictive performance. However, as λ becomes larger, the validation RMSE increases, which suggests that excessive penalization removes too much information from the model and leads to worse prediction accuracy. The minimum validation RMSE occurs at $\lambda = 0.49$, so we select this value as the optimal tuning parameter for the final lasso regression model. Since Salary was divided by 1,000,000 before fitting the model, the reported validation RMSE is measured in millions of U.S. dollars.

Using the optimal tuning parameter $\lambda = 0.49$, we fit the final lasso regression model. Unlike ridge regression, lasso can shrink some coefficient estimates exactly to zero, so it performs variable selection as well as shrinkage. Therefore, the resulting model may be simpler and more interpretable. Table 9 below presents the first 10 coefficient estimates from the fitted lasso model, while the complete coefficient table is provided in the Appendix E at the end of this report. Because Salary was scaled by dividing by 1,000,000, all coefficient estimates are reported in millions of U.S. dollars.

Table 9: First 10 coefficient estimates from the lasso regression model (Salary measured in millions of U.S. dollars)

Variable	Coefficient
(Intercept)	-20.4670
PositionPF	0.0000
PositionPG	0.4291
PositionPG-SG	0.0000
PositionSF	0.0000
PositionSF-PF	0.0000
PositionSF-SG	0.0000
PositionSG	0.0000
PositionSG-PG	0.0000
Age	0.7340

As shown in Table 9, several coefficient estimates are exactly zero under the lasso penalty, indicating that those predictors were excluded from the final model. This confirms that lasso performs variable selection in addition to shrinkage.

Section 4.3: Regression Comparison

In this section, we compare the three regression approaches considered above: the LSLR model, ridge regression, and lasso regression. The LSLR model provides a useful baseline, but it is sensitive to multicollinearity and may produce unstable coefficient estimates when predictors are strongly correlated. Ridge regression addresses this issue by shrinking coefficient estimates toward zero, which improves stability but keeps all predictors in the model. Lasso regression extends this idea by performing both shrinkage and variable selection, allowing some coefficients to be reduced exactly to zero. Therefore, comparing these three methods allows us to evaluate the trade-off between model stability, interpretability, and predictive performance.

Table 10: Comparison of regression models (RMSE measured in millions of U.S. dollars)

Model	Optimal.lambda	Validation.RMSE	Performs.Selection	Number.of.Features.Used
LSLR	NA	NA	No	100
Ridge	3.59	6.9707	No	100
Lasso	0.49	6.6072	Yes	13

Table 10 above compares the regression models in terms of predictive performance and model complexity. Among the penalized regression methods, lasso regression achieves the lowest validation RMSE, indicating the best predictive performance on this dataset. Ridge regression also improves upon the ordinary least squares model by reducing the instability caused by multicollinearity, but it retains all predictors in the model. In contrast, lasso regression not only provides shrinkage, but also removes less important variables by shrinking some coefficients exactly to zero. As a result, lasso produces a simpler and more interpretable model while also achieving the strongest validation performance. Since Salary was divided by 1,000,000 before fitting the models, all RMSE values are interpreted in millions of U.S. dollars.

Overall, lasso regression appears to be the most effective model among those considered, since it achieves the lowest validation RMSE while also producing a more parsimonious model through variable selection. As stated in Table 10 above, given the RMSE for lasso regression equals to (6.6072), meaning on average, the predicted salary of a basketball player is either 6.6072 millions USD higher or lower than the actual salary of that specific player.

Section 5: Decision Tree

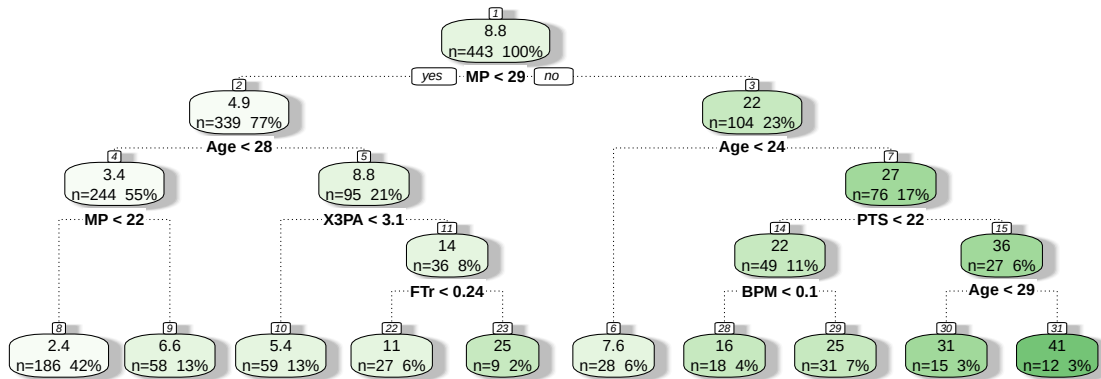
In this section, we approach this problem using a more visualization solution. In addition to the prediction and regression models discussed in the previous sections, we apply tree based methods to better understand how player characteristics relate to salary. Specifically, we first construct a single regression tree to provide an interpretable visual structure of the prediction procedure. Then, we use a random forest model to improve predictive performance by combining multiple trees. Finally, we compare the results from the tree and forest models to evaluate their relative strengths and limitations.

Section 5.1: Tree

Trees are very powerful tools for association and predictive tasks for regression, and our goal is to use all the possible features to build a model to predict $Y = \text{Salary}$ of the basketball players.

The way how regression tree works is that they assume there are clusters in the data defined by the features, and rows in different clusters are predicted to have different values of the response variable (Salary). For example, all players with more than (100) career re-bounces are expected to have an average salary of $\hat{y}_1$ while all players with less than or equals to (100) career re-bounces are expected to have an average salary of $\hat{y}_2$.

The Tree 1 shown below is our tree for predicting player's salary. Out of all features, our tree selected (6) features using for predicting the salary, including 'MP', 'Age', 'X3PA', 'FTr', and 'BPM'. The explanation of the abbreviation is shown in the Table 11 below Tree 1.



Tree 1: Regression Tree for Player Salaries

Table 11: Explanation of Basketball Abbreviations Used in the Regression Tree

Abbreviation	Full Name	Meaning in Basketball
MP	Minutes Played	The average number of minutes a player plays per game or total minutes played, depending on the dataset.
Age	Player Age	The player's age during the season.
X3PA	3-Point Attempts	The number of three-point shots attempted by the player.
FTr	Free Throw Rate	Free throw attempts divided by field goal attempts. It measures how often a player gets to the free-throw line.
PTS	Points	The number of points scored by the player.
BPM	Box Plus/Minus	An advanced metric that estimates a player's overall contribution per 100 possessions compared with a league-average player.

From Tree 1, the first and most important split is based on 'MP', which means minutes played is the strongest variable selected by the tree for predicting player salary. Players with 'MP' less than 29 have a much lower predicted average salary, around 4.9 million dollars, while players with 'MP' greater than or equal to 29 have a much higher predicted average salary, around 22 million dollars. This suggests that playing time is strongly associated with salary, which is reasonable because players who stay on the court longer usually have larger roles on their teams.

For players with lower 'MP', 'Age' and 'X3PA' further divide players into different salary groups. Among players with 'MP' less than 29, younger players and players with fewer minutes played tend to have lower predicted salaries. For example, players with 'MP' less than 22 have an average predicted salary of about 2.4 million dollars, while players with 'MP' between 22 and 29 have a higher predicted salary of about 6.6 million dollars. This shows that even within the lower-playing-time group, players who receive more playing time are still expected to earn more.

For players with higher 'MP', 'Age', 'PTS', and 'BPM' become important splitting variables. Players with 'MP' greater than or equal to 29 and 'Age' greater than or equal to 24 have higher predicted salaries than younger players in the same playing-time group. In addition, players with higher points and higher 'BPM' are predicted to have higher salaries. The highest predicted salary group is players with 'MP' greater than or equal to 29, 'Age' greater than or equal to 29, and 'PTS' greater than or equal to 22, with an average predicted salary of about 41 million dollars. This indicates that veteran players who play many minutes and score many points are usually among the highest-paid players.

Overall, the regression tree suggests that salary is mainly related to playing time first, and then further explained by age, scoring ability, shooting behavior, free throw rate, and overall box-score impact. This makes sense in the basketball context because salary is usually higher for players who have larger roles, stronger offensive production, and better overall performance.

To evaluate the predictive performance of the regression tree, we used the cross-validated root mean squared error (RMSE). Since the response variable Salary was measured in millions of dollars, the RMSE is also interpreted in millions of dollars. The RMSE of the regression tree is (6.72), as shown in Table 12 below, which means that, on average, the predicted salary from the tree either (6.72) million dollars higher or lower than the actual salary of that player. A smaller RMSE would indicate better predictive accuracy. Therefore, this value provides a way to assess how well the tree perform when predicting player salaries on test data.

Table 12: Cross-Validated RMSE for Regression Tree

Metric	Value
Cross-Validated RMSE	6.72

Section 5.2: Forest(s)

Forests are combination of numerous single decision trees. Instead of relying on one tree, forest methods build many trees and combine their predictions. This usually improves prediction accuracy because a single tree can be sensitive to small changes in the data, while a forest averages many trees to reduce instability.

In this section, we fitted two forest based models to predict player salary, including a bagged forest model and a random forest model. The bagged forest model uses all available predictors at each split, while the random forest model uses only a random subset of predictors at each split. In our analysis, both models were built using (500) trees, and the response variable was Salary measured in millions of dollars.

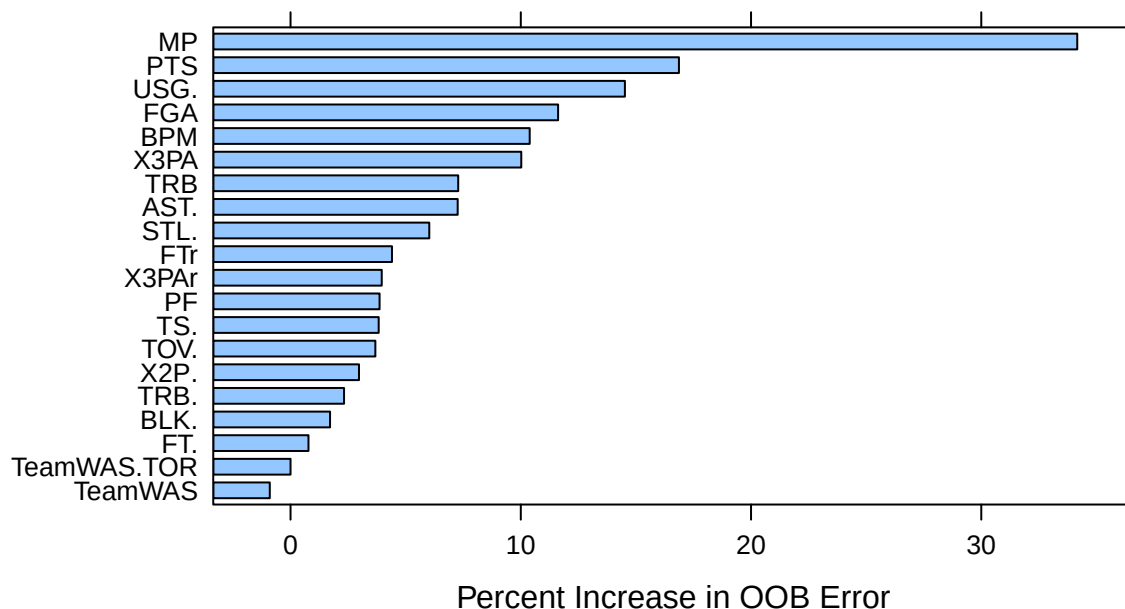
To evaluate the predictive performance of the two forest models, we used the out-of-bag RMSE, as shown in Table 13 below. The bagged forest model had an RMSE of (5.81), while the random forest model had an RMSE of (6.71). Since a smaller RMSE indicates better predictive performance, the bagged forest model performed better than the random forest model in this analysis. This means that the bagged forest model's predicted salary differed from the actual salary by about (5.81) million dollars on average, while the random forest model's predicted salary differed by about (6.71) million dollars on average.

Table 13: Out-of-Bag RMSE for Forest Models

Model	RMSE
Bagged Forest	5.81
Random Forest	6.71

To identify which features are the most important predictors of player salary, we created a feature importance plot for the bagged forest model as shown in Figure 7 below. The plot ranks the predictors based on the percent increase in out-of-bag error. A larger percent increase in out-of-bag error means that the model depends more heavily on that variable, because permuting that predictor causes a larger decrease in predictive accuracy. Therefore, the variables with the highest values in the plot can be interpreted as the most important features for predicting player salary in the bagged forest model.

Figure 7: Importance of Features in Bagged Forest



Based on Figure 7, ‘MP’ is the most important predictor of player salary in the bagged forest model. ‘MP’ has the largest percent increase in out-of-bag error, meaning that the model’s prediction accuracy decreases the most when ‘MP’ is permuted. This suggests that minutes played is the strongest feature for predicting salary.

After ‘MP’, the next most important predictors are ‘PTS’, ‘USG.’, ‘FGA’, ‘BPM’, and ‘X3PA’. These variables are all related to a player’s offensive role and overall involvement in the game. This makes sense because players who play more minutes, score more points, take more shots, and have higher usage rates are often more central to their teams and therefore tend to have higher salaries. Overall, the importance plot suggests that player salary is mainly predicted by playing time, scoring production, and offensive responsibility.

Section 5.3: Comparison

In Sections 5.1 and 5.2, we fitted a regression tree and two forest based models to predict player salary. The single regression tree is easier to interpret because it shows a clear set of decision rules. For example, the tree shows how salary predictions change based on variables such as ‘MP’, ‘Age’, ‘PTS’, and ‘BPM’. This makes the tree useful for understanding the main relationships between player characteristics and salary.

However, the forest models produced better predictive performance than the single tree. As shown in Table 14 below The regression tree had a RMSE of (6.72), while the bagged forest model had an RMSE of (5.81) and the random forest model had an RMSE of (6.71). Since a smaller RMSE indicates better predictive accuracy, the bagged forest model performed the best among the three models. This means the bagged forest model had the lowest average prediction error when predicting player salary.

Table 14: Comparison of Tree and Forest Model Performance

Model	Metric	Value
Regression Tree	Cross-Validated RMSE	6.72
Bagged Forest	Out-of-Bag RMSE	5.81
Random Forest	Out-of-Bag RMSE	6.71

The difference between the tree and forest models shows the trade-off between interpretability and predictive accuracy. The single tree is easier to explain visually, but it may be less stable because one tree can change noticeably if the data changes. In contrast, forest methods combine many trees, which usually makes the model more stable and accurate. Based on the RMSE values, the bagged forest model is the strongest predictive model in this section, while the single regression tree is the most interpretable model.

Section 6: Conclusion

This project examined how NBA player salary can be predicted using measurable performance statistics from the 2022–2023 season. Across the analysis, several important patterns appeared consistently. Minutes played per game was one of the strongest predictors of salary, both in the exploratory analysis and in the tree based models. Players who receive more playing time usually have larger roles on their teams, which helps explain why minutes played is strongly associated with higher salary. Other important variables included points, usage rate, field goal attempts, three-point attempts, age, and BPM, all of which reflect offensive responsibility, experience, and overall player contribution.

The KNN model provided a flexible non-parametric approach and used local similarity among players to predict salary. However, its predictions showed moderate accuracy and tended to underestimate some higher-paid players. The regression models then provided a more structured comparison. Ordinary least squares regression served as a baseline, but multicollinearity made it less reliable. Ridge regression improved stability by shrinking coefficients, while lasso regression improved both prediction and interpretability by selecting a

smaller subset of important variables. Among the regression models, lasso regression performed best, with a validation RMSE of (6.61) million dollars and (13) features retained.

The tree based models offered an additional perspective. The single regression tree was useful because it clearly showed how salary predictions changed based on variables such as 'MP', 'Age', 'PTS', 'X3PA', 'FTr', and 'BPM'. However, the tree was less accurate than the best forest model. The bagged forest model achieved the lowest RMSE among the tree and forest models, with an out-of-bag RMSE of 5.81 million dollars. This made it the strongest predictive model in the final comparison. The random forest model performed similarly to the single tree, with an RMSE of (6.71) million dollars, but did not outperform the bagged forest model in this analysis.

Overall, the results show that player salary can be meaningfully predicted using performance statistics, although salary is not determined by performance alone. Contract timing, team salary cap decisions, player reputation, injuries, market demand, and negotiation factors may also affect salary but are not fully captured in this dataset. Even with these limitations, the models show that playing time, scoring production, offensive usage, and overall impact are strongly related to salary. Based on predictive accuracy, the bagged forest model is the best overall model in this report. Based on interpretability, the single regression tree is the easiest model to explain. Therefore, the final conclusion is that tree based ensemble methods provide the strongest salary predictions, while simpler tree and regression models help explain the main basketball factors associated with NBA player compensation.

Appendix A: The Data

Salary	Age	MP	AST.	TRB	BPM	TOV.
48070014	34	34.7	30.0	6.1	7.5	12.5
47345760	32	22.2	35.3	2.7	-1.2	17.1
47080179	34	29.1	38.6	5.8	0.2	18.4
44474988	38	35.5	33.5	8.3	6.1	11.6
44119845	34	35.6	24.5	6.7	7.1	13.4
43279250	29	33.5	26.6	3.9	1.8	12.9
42492492	31	33.6	19.6	6.5	6.1	8.1
42492492	32	34.6	24.2	6.1	2.8	13.4
42492492	28	32.1	33.2	11.8	8.5	13.2
42492492	32	36.3	35.0	4.8	7.1	11.7
40600080	32	33.0	11.0	4.1	-0.3	8.6
38917057	30	37.4	25.4	5.1	4.1	8.8
38172414	30	30.7	5.6	11.6	0.7	14.9
37984276	31	24.3	30.7	4.2	0.8	13.6
37980720	29	34.0	13.0	12.5	6.3	9.5
37653300	33	33.4	27.1	5.9	8.7	8.2
37633050	30	32.9	11.4	5.7	0.7	9.0
37281261	32	16.0	20.1	1.8	-0.3	5.7
37096500	24	34.8	42.5	3.0	3.3	15.2
37096500	27	35.9	18.7	4.5	1.9	11.0
35448672	26	26.3	31.3	6.3	0.4	27.2
35448672	28	37.4	24.9	7.8	3.1	10.0
35096500	26	29.4	7.5	7.5	2.1	10.8
34319520	32	32.6	34.4	5.1	3.1	15.2
33833400	27	33.0	22.0	8.1	3.0	15.0
33833400	26	34.6	27.3	4.5	4.2	10.6
33616770	27	32.2	10.5	5.0	-0.5	7.9
33616770	28	34.6	22.9	10.2	9.2	11.9
33333333	31	35.3	25.1	4.4	0.8	11.2
33000000	33	36.8	43.3	6.1	5.4	16.3
31650600	25	32.8	27.5	4.0	1.3	11.3
31650600	25	34.2	28.1	5.5	1.7	13.4
31377750	26	32.5	27.4	3.0	1.5	15.1
30913750	26	35.8	21.4	4.3	6.3	10.3
30913750	24	35.5	25.7	4.8	7.3	10.1
30913750	24	30.4	9.2	10.0	0.9	11.0
30556968	34	20.0	13.2	6.4	0.6	13.1
30351780	24	36.9	20.9	8.8	5.5	10.4
30351780	25	34.6	15.9	9.2	1.5	12.7
30351780	25	33.4	29.6	4.2	2.5	10.6
30075000	32	31.5	19.3	4.3	-1.4	13.2
29776785	26	35.9	16.5	6.9	1.3	11.4
28400000	37	32.0	38.7	4.3	3.2	13.4
28333334	36	31.2	23.4	4.1	0.2	15.9
27733332	26	35.0	28.7	3.5	3.9	9.4
27300000	33	36.2	22.3	4.6	2.0	9.1
26500000	36	30.5	13.2	6.2	3.3	7.0
25806468	32	31.5	27.8	7.2	0.8	28.1
23760000	28	35.5	18.5	10.0	3.7	11.5

Salary	Age	MP	AST.	TRB	BPM	TOV.
23500000	25	30.0	5.3	6.5	-1.3	9.1
22680000	35	30.3	29.4	2.7	0.9	13.1
22600000	30	26.0	21.5	4.2	2.8	10.8
22321429	23	35.0	18.6	2.6	-1.1	10.2
21486316	28	35.3	22.7	4.1	-0.6	9.4
21250000	28	36.7	28.1	4.1	2.5	10.2
21177750	30	31.0	13.3	5.0	1.9	11.3
21100000	26	34.6	30.4	12.3	5.8	16.9
20955000	28	35.7	10.3	4.5	-0.2	9.8
20690909	27	30.2	14.2	6.6	2.1	9.8
20100000	26	35.7	13.7	4.4	1.7	8.2
20000000	24	32.6	7.5	9.8	2.4	11.4
19700319	34	21.9	9.4	3.8	0.5	11.1
19568360	34	28.5	13.4	1.9	-1.9	12.5
19500000	29	34.5	28.1	3.4	0.7	10.8
18796296	28	30.2	18.0	3.8	-0.2	12.4
18642857	31	20.6	9.1	2.2	-1.1	8.3
18352273	30	32.5	6.3	4.5	-1.0	8.1
18206896	28	26.6	4.7	11.0	2.2	8.3
18000000	30	17.0	10.6	1.8	-4.0	11.2
17926829	29	27.0	11.2	11.5	0.6	19.8
17400000	25	11.3	7.3	4.0	4.0	6.6
17357143	25	35.6	8.0	5.0	0.4	12.1
17207142	28	32.1	26.4	3.1	-1.0	17.9
17142857	28	28.3	19.4	3.6	2.4	10.1
17071120	26	36.4	24.0	5.3	1.0	10.2
17045454	24	27.0	4.5	9.4	2.4	10.9
16902000	28	16.5	9.7	1.6	-5.0	10.0
16758621	29	26.1	11.0	2.9	0.3	10.8
16700000	24	23.9	17.9	2.2	-1.3	13.3
16500000	24	29.6	29.8	3.9	0.5	15.8
16475454	25	34.4	8.6	8.6	3.8	8.8
16372093	33	28.1	8.9	4.0	-1.9	7.6
15558035	26	25.8	8.4	3.5	-2.0	8.5
15277778	23	30.7	9.2	4.6	-3.0	8.5
15059712	32	17.7	15.4	2.4	-2.6	11.1
15000000	26	24.3	28.8	2.5	1.6	8.9
14520730	34	12.5	19.9	1.5	-3.7	11.8
14508929	24	29.4	14.1	3.3	1.0	9.5
14317459	27	25.9	11.7	7.3	1.7	11.9
14004703	29	31.3	9.8	2.7	-1.0	11.1
13906976	34	30.4	6.1	6.7	2.1	9.8
13801614	34	27.1	13.3	3.7	-1.2	13.4
13750000	31	20.5	9.8	2.2	-3.2	9.5
13745455	26	22.3	9.4	2.7	0.5	10.3
13534817	22	33.0	24.4	7.0	5.8	14.6
13340000	30	32.6	20.9	4.0	-1.4	14.0
13000000	28	24.7	6.9	2.0	-0.7	8.0
13000000	27	35.0	16.5	7.2	-1.7	13.2
12960000	27	32.3	16.1	7.8	0.9	16.4
12939848	29	30.4	6.7	4.8	-2.0	10.5
12804878	31	28.6	17.6	6.2	0.4	20.3

Salary	Age	MP	AST.	TRB	BPM	TOV.
12690000	30	20.0	2.7	5.5	-0.1	6.4
12307692	32	16.2	10.2	3.5	1.7	11.5
12196094	29	23.5	15.8	2.7	-1.2	12.7
12119440	23	31.9	41.1	5.9	5.7	12.6
11710528	35	12.5	6.1	1.3	-1.9	16.8
11500000	31	21.1	10.2	2.4	-2.1	8.2
11457217	32	24.4	20.0	2.1	-3.9	14.6
11400000	27	30.3	11.7	3.3	-3.9	8.9
11215260	29	8.3	3.5	1.9	-2.9	12.6
11080125	31	19.2	6.9	4.1	-1.2	15.9
11055120	20	33.8	17.2	6.9	-1.5	12.8
10900634	22	33.9	12.4	5.0	-3.1	10.8
10843350	27	26.0	8.7	9.6	0.7	8.6
10733758	34	15.0	7.7	5.2	-1.8	26.1
10733400	21	36.0	19.3	5.8	1.0	13.0
10714287	25	23.5	8.5	8.3	3.3	15.3
10552800	21	33.3	30.7	6.2	-0.7	13.8
10300000	24	15.7	8.7	4.6	0.5	10.1
10260000	22	20.2	27.4	3.2	-1.6	15.5
10183800	32	18.9	10.6	3.8	1.2	6.6
10123457	25	28.6	5.1	9.9	-0.9	15.9
10012800	31	30.3	6.2	3.6	-2.6	6.9
10012800	31	22.0	15.9	3.1	1.2	9.7
9835881	25	31.7	6.0	4.2	-3.3	8.2
9603360	21	19.3	5.5	5.9	-3.5	11.8
9500000	25	20.2	15.7	1.7	-1.4	10.0
9472219	24	22.3	25.0	2.6	0.3	14.1
9441840	20	34.2	17.3	3.7	-2.1	11.3
9398148	27	26.5	15.2	9.1	1.9	15.0
9391069	28	11.5	6.7	2.7	-3.8	23.9
9200000	29	31.7	15.3	5.1	-1.1	15.5
9125000	27	27.3	26.4	3.4	0.5	9.9
9080417	32	26.0	17.3	8.9	2.2	15.8
9030000	28	23.5	15.4	2.9	1.1	19.5
9000000	31	25.1	7.5	3.6	-2.2	12.7
8925000	27	27.4	10.9	3.3	0.1	10.5
8920794	23	35.5	34.1	2.7	2.4	13.5
8854879	24	19.8	8.3	6.5	-0.3	15.6
8780488	29	28.4	22.6	5.3	2.5	15.7
8750000	30	26.3	19.8	3.0	-2.1	16.6
8694369	30	15.6	12.2	3.1	-1.4	5.7
8623920	21	35.2	38.7	6.4	2.4	14.3
8496653	32	19.9	11.9	1.6	-2.1	9.7
8478720	21	34.4	12.4	9.0	1.7	11.9
8300000	30	16.7	11.0	3.1	0.3	18.2
8283688	33	11.2	9.5	3.5	-1.6	12.7
8250000	24	27.9	12.9	4.1	0.9	12.9
8200000	32	14.0	8.9	3.6	-0.5	15.6
8008440	22	29.8	5.5	4.6	-0.6	6.8
8000000	26	23.9	13.8	9.3	2.1	9.1
8000000	34	14.7	12.9	3.1	0.1	16.7
7804878	30	24.4	20.9	3.6	3.4	12.2

Salary	Age	MP	AST.	TRB	BPM	TOV.
7775400	21	28.3	5.8	4.0	-1.3	12.3
7644600	21	34.8	20.0	6.6	0.4	12.0
7426088	27	20.1	10.6	2.3	-0.7	8.3
7413955	22	23.4	16.3	2.9	-0.8	10.6
7350000	25	22.9	18.0	6.4	-0.2	17.4
7252200	20	31.1	26.2	3.9	-3.3	17.2
7100000	28	22.1	9.6	2.4	-3.3	14.3
7065217	27	21.8	18.2	3.8	-3.5	15.3
7040880	22	21.7	7.0	2.5	-0.3	9.5
7000000	27	19.1	11.0	3.4	-4.6	4.9
6922320	21	23.5	18.0	3.0	-1.0	15.9
6803012	22	13.0	7.7	2.8	-1.2	13.9
6667500	30	8.1	5.8	1.3	-3.5	19.0
6586800	20	28.5	7.6	4.1	-3.7	11.7
6479000	26	28.5	15.7	4.1	-0.5	13.2
6479000	27	29.3	8.2	4.8	-1.2	11.5
6479000	35	22.7	18.4	2.8	-0.7	17.6
6395160	22	23.1	5.8	7.2	0.8	12.2
6287400	20	31.1	29.9	7.9	1.2	15.1
6263188	24	23.3	5.8	4.5	-2.2	7.7
6184500	36	14.6	9.4	2.9	-4.0	10.7
6025000	27	22.3	14.5	2.5	-0.6	11.8
6012960	19	22.2	7.9	3.0	-3.3	10.7
6000000	28	20.2	32.6	2.2	-1.2	14.4
5987975	33	15.3	7.5	1.2	-3.9	9.6
5954454	23	24.8	8.1	2.2	-3.4	12.0
5887899	26	28.5	9.6	4.4	2.6	6.6
5853659	32	17.2	27.1	2.1	-2.8	13.8
5837760	21	28.3	32.3	2.9	-3.8	16.7
5739840	20	20.8	12.7	3.4	-1.7	14.6
5728393	30	23.7	6.9	4.6	-1.0	7.0
5722116	23	34.9	19.9	5.4	0.6	11.7
5634257	23	19.6	7.9	2.7	-4.2	9.6
5508600	19	17.7	16.4	3.2	-1.8	20.2
5461219	35	8.5	5.2	2.5	-3.7	20.1
5377520	23	12.0	12.1	3.6	0.3	13.3
5348280	24	15.7	9.2	2.8	-0.5	8.0
5258280	21	32.6	17.2	4.1	-0.1	11.7
5250000	27	13.8	13.8	1.6	-3.5	12.4
5200000	24	17.8	9.2	3.5	-1.4	13.9
5155500	31	19.8	25.4	1.7	-1.5	12.8
5155500	36	6.5	10.6	0.7	-1.0	7.9
5121951	32	24.7	7.9	5.4	-0.6	12.3
5063520	19	26.0	13.6	5.3	-4.0	13.9
5009633	24	15.0	16.7	1.7	-0.9	14.6
5000000	25	9.5	8.8	1.1	-4.4	15.9
4916160	22	26.6	13.9	6.4	-2.3	15.9
4833600	24	18.1	16.5	1.3	-2.8	13.1
4810320	20	15.1	9.2	2.3	-7.4	8.5
4728948	25	18.1	22.1	1.7	-0.8	10.3
4670160	22	18.8	7.6	5.8	-2.1	11.2
4591680	21	15.2	8.2	2.1	-5.6	15.2

Salary	Age	MP	AST.	TRB	BPM	TOV.
4569840	19	14.6	10.7	2.7	-3.6	12.0
4500000	36	19.5	8.3	2.6	-3.4	10.8
4500000	26	26.3	17.3	4.5	0.7	16.4
4437000	22	31.0	17.3	3.9	0.4	8.2
4379526	25	16.9	5.9	2.0	-0.5	8.3
4374000	23	24.1	13.1	7.5	1.1	15.9
4362240	22	15.1	11.1	2.1	-6.5	13.4
4343919	26	19.5	9.5	5.5	2.5	11.8
4341480	21	30.3	15.4	4.5	0.3	12.3
4306281	24	25.9	8.4	4.6	-1.1	13.4
4264628	22	14.3	8.3	3.2	-1.0	11.2
4215120	22	33.6	47.6	3.7	7.2	13.2
4171548	22	18.1	7.3	2.6	-3.0	10.0
4124280	19	24.9	6.8	8.9	-0.8	16.5
4105236	33	11.5	19.4	3.5	-3.9	20.7
4097561	26	26.8	17.2	5.0	-2.8	17.0
4037277	26	3.3	15.2	0.0	6.7	25.0
4022164	26	15.6	19.0	3.2	-1.0	19.2
4000000	36	18.1	16.7	1.8	-1.5	15.2
4000000	25	13.1	11.1	2.2	-1.5	12.2
3936960	25	19.5	9.7	2.5	-4.5	10.1
3918600	29	6.2	9.3	2.3	-4.0	30.2
3918360	22	20.5	7.5	2.1	-3.0	9.0
3901399	23	30.0	22.5	2.7	-1.9	14.7
3873024	23	32.7	13.7	5.0	-1.8	9.6
3841067	35	9.0	12.6	1.7	-4.4	13.2
3804360	23	24.9	7.4	3.8	-3.0	10.3
3740160	20	13.0	8.2	1.7	-2.6	12.0
3722040	21	19.3	3.3	7.1	0.1	12.0
3613680	22	25.9	22.9	4.8	1.3	11.7
3552840	23	28.3	5.7	2.8	-1.5	8.1
3536160	19	19.5	7.0	2.1	-0.8	7.1
3500000	31	15.0	7.1	3.2	-0.4	6.8
3465000	29	19.4	7.3	2.4	-1.4	9.4
3433320	21	28.3	7.3	8.1	-2.1	11.8
3433320	24	19.2	9.4	3.6	-1.6	11.5
3395062	25	12.5	11.2	1.1	-5.0	10.8
3359160	21	21.5	7.4	6.0	-1.7	12.2
3261480	21	20.6	12.4	4.7	0.0	14.0
3205128	27	14.7	7.5	1.7	-5.2	10.5
3200000	29	12.7	5.7	6.6	-0.8	18.6
3191400	20	5.6	14.1	1.0	0.0	9.0
3122602	27	6.0	9.4	2.0	-3.3	0.0
3098400	22	25.7	9.7	3.0	-2.4	14.5
3047640	21	11.8	6.2	1.8	-3.5	15.4
3046200	21	17.7	14.0	2.3	-3.9	10.6
3000000	23	15.1	5.2	2.8	-3.5	12.2
2959080	23	27.6	8.1	4.7	-0.3	7.0
2925600	19	23.5	11.4	2.7	-5.2	10.6
2909261	23	5.6	7.4	0.4	-10.6	6.7
2909040	22	12.0	3.7	2.7	-4.0	19.1
2905851	33	13.9	14.2	3.8	0.1	13.2

Salary	Age	MP	AST.	TRB	BPM	TOV.
2905851	34	8.1	9.1	1.4	-3.0	23.2
2905851	33	10.1	12.2	2.0	-2.7	13.1
2905851	39	14.1	20.3	2.1	-2.5	35.5
2905851	33	11.6	2.9	2.8	-4.5	15.6
2905851	42	10.1	0.0	1.6	-12.1	3.1
2905851	36	15.8	5.1	2.2	-2.7	9.5
2905851	30	19.5	9.3	1.6	-2.8	9.2
2905851	30	14.3	8.7	4.3	-0.9	17.7
2905851	37	9.8	9.6	1.9	-3.2	15.3
2840160	23	20.7	6.3	6.0	-2.3	11.4
2808600	21	15.5	6.4	2.4	-2.4	9.9
2800000	24	19.2	10.5	3.7	-2.1	12.0
2792640	21	14.9	10.4	4.0	0.3	10.3
2726880	22	33.6	16.9	2.9	0.6	7.4
2717391	24	10.4	10.8	3.5	-2.0	15.7
2696400	21	23.0	5.7	8.4	2.9	10.7
2681040	20	10.4	20.9	1.1	-5.4	18.0
2674742	22	14.5	12.9	5.5	0.6	21.3
2628597	32	6.7	23.4	0.4	-5.2	15.9
2625000	27	16.9	7.0	4.1	-1.0	12.2
2617800	22	13.7	3.3	2.6	-3.7	12.2
2588640	21	18.0	6.1	2.8	-3.9	11.3
2573760	21	16.5	7.0	4.5	-0.8	11.8
2485200	22	13.5	6.7	2.2	-4.8	14.8
2471160	20	12.9	8.8	4.1	-1.4	19.2
2463490	30	10.5	21.0	1.0	-0.2	16.1
2463490	29	6.5	17.8	1.4	2.8	11.4
2463490	29	11.9	7.4	2.8	-1.3	12.1
2385480	19	18.1	18.8	2.2	-9.6	21.7
2372160	21	12.3	14.3	1.1	-4.3	14.4
2330681	24	11.3	10.7	2.5	-4.6	15.2
2316876	25	13.0	7.0	1.8	0.1	5.1
2316240	23	28.9	17.4	4.2	1.7	8.7
2300000	26	20.8	8.2	4.3	0.5	9.9
2298385	31	16.4	7.6	2.3	-1.5	6.4
2298385	29	10.7	8.2	0.6	-6.3	11.8
2298385	27	11.6	4.1	3.0	-1.7	18.9
2277000	22	29.9	9.7	3.2	-0.1	9.7
2266640	23	14.9	24.3	1.8	-1.8	14.1
2244111	23	27.6	22.4	2.7	-3.2	15.8
2239200	25	13.4	14.0	1.8	-3.4	13.7
2201520	22	19.4	23.4	2.4	-0.8	11.7
2200000	23	21.5	6.9	5.8	-0.9	17.6
2193960	20	8.1	7.9	1.6	-2.8	10.6
2174880	23	10.0	4.3	3.3	-0.6	18.2
2161440	22	30.6	8.4	3.9	-1.2	12.6
2160000	26	15.8	27.4	1.4	-0.7	17.7
2145720	24	13.0	13.7	1.4	-4.1	8.9
2138160	21	16.6	12.7	1.7	-2.7	10.6
2133278	27	11.7	9.1	2.9	1.0	11.8
2133278	30	20.4	8.9	3.0	-1.7	13.5
2133278	29	13.5	11.8	2.4	-3.8	18.1

Salary	Age	MP	AST.	TRB	BPM	TOV.
2130240	24	31.7	20.8	5.0	3.5	10.9
2125200	20	5.6	13.8	0.9	0.3	7.2
2109480	21	11.5	9.7	4.2	-2.3	17.7
2036318	25	2.8	4.4	0.9	4.9	6.8
2036318	24	12.9	12.8	1.3	-6.7	16.5
2000000	26	12.7	5.1	1.4	-2.4	7.8
2000000	20	18.7	10.7	4.9	-1.7	12.6
2000000	22	18.9	7.0	4.2	-1.5	6.7
2000000	28	22.8	11.7	4.9	0.5	8.1
2000000	20	15.9	4.9	1.9	-4.0	8.7
1997718	26	20.6	22.3	2.5	-2.0	14.5
1968175	26	13.4	13.0	1.2	-2.0	13.3
1968175	28	16.0	7.0	2.4	-0.6	8.0
1968175	27	13.9	13.3	1.5	-2.6	13.6
1968175	27	8.1	19.6	1.6	-3.4	11.5
1968175	25	18.3	4.3	5.7	0.6	8.2
1968175	24	18.8	10.3	3.5	-0.2	11.2
1968175	25	20.3	9.2	5.4	0.1	14.7
1930681	26	20.3	5.4	3.4	-0.7	8.3
1930681	26	23.1	14.0	3.4	-0.2	12.2
1930681	23	19.3	15.9	2.6	-2.5	8.9
1930681	23	18.4	9.6	4.9	1.0	12.7
1930681	25	23.9	9.6	4.3	-2.1	12.1
1930681	24	20.6	7.5	5.6	1.0	15.1
1910860	25	10.4	8.1	3.0	-0.1	11.3
1902133	27	8.0	8.2	1.4	-5.3	17.5
1902133	23	7.1	6.3	1.0	-3.2	14.4
1878720	25	15.1	4.8	4.2	-2.2	13.4
1878720	30	5.6	9.3	1.2	-7.6	17.7
1878720	25	19.5	11.5	4.5	0.6	12.5
1878720	27	21.7	9.8	3.7	-0.6	9.5
1846738	24	16.7	5.4	3.4	-3.2	8.2
1836090	25	7.7	6.1	1.5	-3.2	16.5
1836090	27	4.7	10.5	0.7	-2.6	6.7
1836090	23	19.1	8.7	2.4	0.9	6.6
1836090	28	8.6	16.9	0.8	-8.8	15.9
1836090	30	10.6	7.2	1.7	-3.3	8.5
1815677	29	15.0	6.4	2.8	1.4	13.5
1815677	26	25.9	13.9	2.1	-3.4	13.3
1815677	26	28.4	11.2	3.2	-1.7	7.8
1782621	25	18.7	4.5	6.4	-1.3	13.0
1782621	25	18.1	4.0	3.3	-2.8	7.9
1782621	25	23.3	14.4	3.6	-1.9	13.0
1782621	23	10.9	5.3	3.8	-0.1	16.3
1782621	23	29.2	31.5	3.6	-1.0	11.8
1752638	26	17.9	6.2	3.5	-3.1	16.5
1752638	24	9.2	3.9	2.6	-3.9	11.9
1709300	22	4.9	12.3	0.5	-14.7	25.9
1703902	23	8.5	5.1	1.5	-4.1	10.1
1674655	22	12.8	15.0	0.7	-2.6	13.5
1639842	24	15.6	4.0	4.7	-0.5	19.1
1637966	25	16.1	7.6	2.6	1.0	6.9

Salary	Age	MP	AST.	TRB	BPM	TOV.
1637966	22	10.7	7.5	1.7	-3.9	11.5
1575000	21	10.2	7.9	2.0	-14.2	9.1
1563518	22	5.7	12.4	0.9	-3.6	11.7
1563518	20	14.0	4.9	2.2	-4.7	14.5
1563518	23	2.9	0.0	0.3	-14.4	0.0
1563518	23	26.2	13.2	2.8	-2.5	14.0
1563518	24	23.1	4.8	2.8	-2.4	9.0
1563518	20	16.0	19.2	1.7	-6.5	25.4
1563518	23	8.9	30.9	1.6	-2.6	19.3
1563518	24	28.8	15.7	3.0	0.7	14.0
1563518	24	21.5	19.9	2.3	-1.2	13.4
1563518	22	11.9	12.4	0.8	-2.3	9.7
1563518	24	18.5	8.3	3.0	-2.1	12.2
1563518	27	14.2	10.1	4.1	-1.1	13.3
1563518	22	19.1	16.0	3.8	-1.8	16.2
1563518	23	9.0	19.3	1.5	-0.5	11.4
1500000	22	13.8	5.0	2.9	-3.5	10.2
1461055	20	17.1	9.1	2.0	-6.0	15.3
1280206	24	17.8	20.8	3.3	0.7	12.9
1200000	20	7.8	26.4	1.1	-2.7	16.9
1160955	27	7.4	5.1	2.1	-6.3	30.7
1116112	23	15.3	12.8	4.3	-2.4	12.1
1105522	26	11.8	12.0	1.8	1.2	0.0
1017781	23	4.1	3.9	0.8	-7.4	7.7
1017781	20	14.8	14.9	1.9	-2.3	11.1
1017781	20	5.2	12.4	1.0	-15.0	41.4
1017781	19	12.5	5.1	1.8	-3.1	10.3
1017781	20	6.4	7.5	1.7	-1.6	7.1
1017781	20	11.1	7.8	2.3	-6.1	11.8
1013119	25	16.1	6.3	2.8	-4.4	11.5
1000001	28	25.8	32.3	4.5	2.4	12.8
877940	25	12.2	9.6	1.7	-2.9	10.8
853241	21	6.8	16.3	1.6	-7.9	0.0
838891	25	10.0	11.1	2.4	-7.2	18.3
824041	23	8.2	2.1	3.9	-0.6	10.5
806097	25	9.1	6.9	0.8	-3.4	5.8
784654	26	4.9	16.3	0.9	-7.8	13.2
743922	30	14.5	7.2	4.3	-2.8	15.8
713558	30	12.7	1.2	3.8	-5.0	15.4
694878	25	19.3	10.5	3.5	-1.1	13.7
652829	31	11.0	22.6	1.3	-6.3	19.0
612162	25	8.0	13.9	0.7	-2.5	9.7
520589	19	2.7	0.0	0.7	-13.0	0.0
511816	22	2.4	0.0	0.2	-11.3	0.0
508891	24	12.0	16.5	1.6	-5.1	19.7
508891	25	9.0	0.0	2.5	-8.9	11.3
508891	20	5.5	22.4	0.8	-4.2	16.8
508891	27	1.8	0.0	0.6	2.4	0.0
508891	26	8.0	6.3	1.4	-3.4	19.1
508891	23	7.0	5.0	1.7	-1.4	3.3
508891	25	3.9	7.7	1.0	-1.9	9.0
508891	23	9.0	5.8	0.6	-9.5	0.0

Salary	Age	MP	AST.	TRB	BPM	TOV.
508891	23	6.3	6.2	1.5	3.5	0.0
508891	23	5.6	13.9	0.0	-8.4	8.8
508891	19	6.7	9.7	1.0	-2.9	20.2
508891	23	5.9	12.0	1.0	-3.4	0.0
508891	23	13.3	7.6	2.9	0.8	7.3
508891	24	8.7	11.8	2.3	-0.4	9.2
508891	25	13.4	4.7	2.1	-3.6	14.4
508891	23	4.4	10.5	0.5	1.3	5.0
508891	23	5.8	5.0	2.2	-3.4	9.2
508891	19	14.6	7.6	3.6	-4.6	13.4
508891	21	18.8	4.5	3.5	-3.3	7.4
508891	25	10.4	15.5	0.9	-1.9	7.6
508891	24	7.4	11.2	2.3	3.5	11.8
508891	21	12.9	4.7	2.2	-6.5	2.8
503041	20	11.6	6.5	0.9	-6.2	1.8
500000	27	4.0	16.1	1.0	-20.7	41.0
464299	31	6.5	22.6	0.3	-1.2	17.8
386055	22	13.7	8.4	4.1	-2.1	13.5
374357	25	10.5	19.3	1.6	-0.9	6.1
307089	23	14.1	2.8	2.6	-6.1	6.0
307089	23	29.0	12.2	4.0	2.3	13.0
260295	23	13.7	5.5	3.8	-4.2	12.6
257371	23	12.7	4.6	3.4	-4.9	10.8
254446	23	5.0	0.0	1.0	-26.5	100.0
239822	21	2.0	0.0	1.0	-15.2	0.0
219630	24	15.0	15.1	0.9	-5.1	7.2
211045	26	3.0	0.0	0.0	-22.6	50.0
201802	26	2.7	0.0	0.7	17.8	0.0
201802	25	3.0	0.0	0.3	2.9	0.0
193027	21	9.2	17.0	1.3	-7.6	18.0
192293	26	7.5	4.2	2.5	-9.1	36.8
169630	23	8.8	17.6	1.7	-5.3	23.1
160856	24	20.5	35.3	5.0	0.5	11.9
140384	27	12.6	6.4	3.8	-3.6	19.8
134862	29	24.0	34.9	4.4	0.1	12.3
129405	22	4.5	14.2	0.8	-10.2	39.6
116986	28	5.0	35.7	0.8	-5.1	10.0
116986	23	14.2	7.8	1.2	-4.2	21.2
116986	24	13.6	14.7	3.0	4.0	13.5
116574	25	31.5	40.0	3.2	0.0	11.9
113114	24	5.0	22.7	2.0	-22.2	0.0
105522	29	3.0	0.0	0.0	-24.2	0.0
105522	26	9.8	11.4	1.5	3.5	11.8
102910	25	12.0	13.9	1.3	-3.6	20.3
99438	24	12.4	23.8	1.0	-6.7	22.7
96514	24	3.0	23.5	0.0	17.0	0.0
58493	23	2.0	0.0	0.0	48.6	0.0
52644	23	25.4	12.1	3.0	-3.0	9.0
49719	22	15.0	5.0	3.0	-1.9	27.6
35096	23	22.3	5.9	3.8	-9.0	14.6
32795	25	5.6	21.8	0.8	-0.9	12.7
32171	29	18.7	12.3	2.0	-3.5	0.0

Salary	Age	MP	AST.	TRB	BPM	TOV.
5849	23	35.0	29.0	9.0	-2.0	23.7
5849	24	41.0	19.4	4.0	-6.1	40.0

Appendix B: Revisions

Part 1

Section 1

No Addressing Needed.

Section 2

No Addressing Needed.

Section 3

Comment Received: *‘Nice! My only comment here is that in the final figure on Page 12, it would help to put the X and Y axes on the same range. This makes it easier to compare the truth and predictions. Let me know if you are not sure how to do this!’*

Addressing: Fixed the range for both X and Y axis. See Figure 4 for the revised plot with equal x- and y-axis ranges.

Part 2

Section 4

No Addressing Needed.

Appendix C: Full coefficient summary for the LSLR

Table 16: Full coefficient summary for the LSLR model

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3937945.46	8977907.05	-0.4386	0.6612
PositionPF	-1035172.43	1363116.21	-0.7594	0.4481
PositionPG	1890715.87	2089278.61	0.9050	0.3661
PositionPG-SG	4439934.88	6290111.50	0.7059	0.4808
PositionSF	1342520.55	1588642.79	0.8451	0.3987
PositionSF-PF	736689.51	6883075.00	0.1070	0.9148
PositionSF-SG	-195446.32	8175409.28	-0.0239	0.9809
PositionSG	-186726.67	1750160.18	-0.1067	0.9151
PositionSG-PG	221003.93	8057332.72	0.0274	0.9781
Age	798813.10	90883.71	8.7894	0.0000
TeamBOS	238577.35	2496859.13	0.0956	0.9239

Variable	Estimate	Std. Error	t value	Pr(> t)
TeamBOS/OKC	-4182299.18	6901685.41	-0.6060	0.5449
TeamBRK	1000745.68	2968493.41	0.3371	0.7362
TeamBRK/DAL	1719749.58	4749756.00	0.3621	0.7175
TeamBRK/DET	872462.02	7132098.70	0.1223	0.9027
TeamBRK/LAC	-1258129.25	7258486.12	-0.1733	0.8625
TeamBRK/MIA	-1877979.35	6958863.36	-0.2699	0.7874
TeamBRK/PHO	-1192234.13	5151644.71	-0.2314	0.8171
TeamBRK/SAC	1318680.62	6785436.70	0.1943	0.8460
TeamCHI	229006.15	2683557.83	0.0853	0.9320
TeamCHI/IND	2930015.47	6831027.53	0.4289	0.6682
TeamCHO	-529079.95	2692956.65	-0.1965	0.8444
TeamCLE	-2061392.41	2516936.74	-0.8190	0.4134
TeamCLE/MEM	1717471.88	7044249.25	0.2438	0.8075
TeamCLE/MIA	16963770.94	7029503.59	2.4132	0.0163
TeamDAL	1164393.14	2817238.99	0.4133	0.6796
TeamDAL/ATL	-4192986.16	6877634.43	-0.6097	0.5425
TeamDEN	902308.39	2818297.92	0.3202	0.7490
TeamDEN/LAC	-5193029.20	5065139.88	-1.0252	0.3060
TeamDEN/LAL	-4020776.51	4985098.77	-0.8066	0.4205
TeamDET	-2768839.26	2658974.91	-1.0413	0.2985
TeamDET/ATL	-10649574.17	6788470.41	-1.5688	0.1176
TeamDET/OKC	-5423767.57	6827241.04	-0.7944	0.4275
TeamGSW	2600519.48	2640808.58	0.9847	0.3254
TeamGSW/DET	3248664.90	6800969.68	0.4777	0.6332
TeamGSW/POR	3291981.80	6947165.27	0.4739	0.6359
TeamHOU	-1091905.50	2976807.08	-0.3668	0.7140
TeamHOU/ATL	-433093.26	4315924.43	-0.1003	0.9201
TeamHOU/LAC	2326721.35	6921830.60	0.3361	0.7370
TeamIND	-2236783.08	2594444.74	-0.8621	0.3892
TeamLAC	4846097.89	2761151.70	1.7551	0.0801
TeamLAC/CHO	-9038310.06	6917975.53	-1.3065	0.1923
TeamLAC/MEM	8033096.11	6873014.81	1.1688	0.2433
TeamLAL	-375810.38	3505072.35	-0.1072	0.9147
TeamLAL/CHI	2207534.64	10418897.57	0.2119	0.8323
TeamLAL/LAC	22094096.09	6938201.30	3.1844	0.0016
TeamLAL/MIN	6315078.02	5012180.56	1.2599	0.2085
TeamLAL/ORL	6112258.19	6892551.17	0.8868	0.3758
TeamLAL/UTA	-949548.19	3756115.71	-0.2528	0.8006
TeamLAL/WAS	-2759898.68	5021403.39	-0.5496	0.5829
TeamMEM	-3289458.86	2696994.96	-1.2197	0.2234
TeamMIA	-1673230.20	2573031.51	-0.6503	0.5159
TeamMIA/WAS	-3669226.52	6937969.47	-0.5289	0.5972
TeamMIL	-1674343.74	2554033.21	-0.6556	0.5125
TeamMIL/IND	-8470412.72	5854433.70	-1.4468	0.1489
TeamMIN	-940223.36	2596845.54	-0.3621	0.7175
TeamMIN/UTA	3000359.71	5047797.76	0.5944	0.5526
TeamNOP	-1160677.39	2966421.11	-0.3913	0.6958
TeamNOP/SAS	862859.86	6801283.39	0.1269	0.8991
TeamNYK	-22105.95	2649319.92	-0.0083	0.9933
TeamNYK/POR	-1171247.21	5050431.09	-0.2319	0.8167
TeamOKC	-1602815.86	2565635.33	-0.6247	0.5326
TeamOKC/PHO	-572971.58	6921854.21	-0.0828	0.9341

Variable	Estimate	Std. Error	t value	Pr(> t)
TeamORL	-1795229.69	2537004.21	-0.7076	0.4797
TeamORL/IND	2799224.58	6857346.01	0.4082	0.6834
TeamORL/PHO	722918.28	6796172.41	0.1064	0.9153
TeamPHI	-1840827.80	2709600.95	-0.6794	0.4974
TeamPHI/CHO	-7831601.26	6790434.07	-1.1533	0.2496
TeamPHI/MIA	-1315094.62	6932895.52	-0.1897	0.8497
TeamPHI/PHO	-3792138.08	6834613.03	-0.5548	0.5794
TeamPHI/POR	1272176.87	10581830.03	0.1202	0.9044
TeamPHO	-1438877.84	2829019.15	-0.5086	0.6114
TeamPOR	-1161551.07	2633554.25	-0.4411	0.6594
TeamSAC	947095.50	2511107.93	0.3772	0.7063
TeamSAS	-2636569.58	2471374.43	-1.0668	0.2868
TeamSAS/MIL	-2944590.07	6858949.22	-0.4293	0.6680
TeamSAS/PHI	-3915738.67	6851238.68	-0.5715	0.5680
TeamSAS/TOR	-7514128.90	6818509.24	-1.1020	0.2712
TeamTOR	-1457518.01	2662191.08	-0.5475	0.5844
TeamUTA	-3454635.69	2628418.15	-1.3143	0.1896
TeamWAS	-2219509.71	2531412.26	-0.8768	0.3812
TeamWAS/TOR	3476396.08	6823075.72	0.5095	0.6107
MP	-35260.92	181604.72	-0.1942	0.8462
FGA	-145420.32	730096.52	-0.1992	0.8422
X3PA	-83791.22	484777.42	-0.1728	0.8629
X2P.	631593.93	4402795.16	0.1435	0.8860
FT.	852855.86	2624523.38	0.3250	0.7454
TRB	973492.88	544635.41	1.7874	0.0748
PF	-1125632.69	813442.44	-1.3838	0.1673
PTS	1096684.82	447829.79	2.4489	0.0148
TS.	-25280004.10	10300368.91	-2.4543	0.0146
X3PAr	-4002770.52	3809027.87	-1.0509	0.2941
FTr	1744548.02	2954905.61	0.5904	0.5553
TRB.	-278943.37	238124.95	-1.1714	0.2422
AST.	-115200.61	98444.85	-1.1702	0.2427
STL.	160599.93	668401.67	0.2403	0.8103
BLK.	183948.67	347086.98	0.5300	0.5965
TOV.	88348.96	103812.06	0.8510	0.3953
USG.	19110.54	194062.77	0.0985	0.9216
BPM	473320.23	341760.94	1.3849	0.1670

Appendix D: Full coefficient summary for the ridge regression model

Table 17: Full coefficient summary for the ridge regression model

Variable	Coefficient
(Intercept)	-14.8107
PositionPF	-0.4325
PositionPG	0.7521
PositionPG-SG	3.8449
PositionSF	0.7571

Variable	Coefficient
PositionSF-PF	0.0520
PositionSF-SG	0.0238
PositionSG	-0.7370
PositionSG-PG	0.0552
Age	0.5700
TeamBOS	1.1563
TeamBOS/OKC	-2.9186
TeamBRK	0.2487
TeamBRK/DAL	1.5241
TeamBRK/DET	3.5101
TeamBRK/LAC	-2.4311
TeamBRK/MIA	-1.6182
TeamBRK/PHO	1.5280
TeamBRK/SAC	1.6164
TeamCHI	0.7395
TeamCHI/IND	2.4333
TeamCHO	-0.1218
TeamCLE	-0.8506
TeamCLE/MEM	2.9607
TeamCLE/MIA	12.4992
TeamDAL	1.2931
TeamDAL/ATL	-0.8752
TeamDEN	1.3591
TeamDEN/LAC	-4.2446
TeamDEN/LAL	-1.9641
TeamDET	-2.3988
TeamDET/ATL	-7.1523
TeamDET/OKC	-3.6895
TeamDET/POR	0.0521
TeamGSW	2.2369
TeamGSW/DET	2.2096
TeamGSW/POR	2.8487
TeamHOU	-0.8545
TeamHOU/ATL	-0.6161
TeamHOU/LAC	4.1168
TeamIND	-1.3878
TeamLAC	4.8231
TeamLAC/CHO	-5.2354
TeamLAC/MEM	5.8085
TeamLAL	1.9687
TeamLAL/CHI	2.1937
TeamLAL/LAC	17.3005
TeamLAL/MIN	4.9894
TeamLAL/ORL	3.5119
TeamLAL/UTA	-0.1577
TeamLAL/WAS	-1.4442
TeamMEM	-1.8695
TeamMIA	-0.7384
TeamMIA/WAS	-0.2294
TeamMIL	-0.0075
TeamMIL/IND	-5.2234
TeamMIN	-0.5193

Variable	Coefficient
TeamMIN/UTA	2.3638
TeamNOP	0.2926
TeamNOP/SAS	1.2547
TeamNYK	0.6159
TeamNYK/POR	-0.3354
TeamOKC	-1.2663
TeamOKC/PHO	0.1674
TeamORL	-1.2736
TeamORL/IND	1.3595
TeamORL/PHO	1.7255
TeamPHI	-1.1576
TeamPHI/CHO	-5.6222
TeamPHI/MIA	-1.0792
TeamPHI/PHO	-3.0736
TeamPHI/POR	1.2963
TeamPHO	-0.3182
TeamPOR	-0.9769
TeamSAC	1.0070
TeamSAS	-2.0559
TeamSAS/MIL	-2.8627
TeamSAS/PHI	-3.4788
TeamSAS/TOR	-4.2172
TeamTOR	-0.2807
TeamUTA	-2.1283
TeamWAS	-1.2971
TeamWAS/TOR	3.9053
MP	0.1224
FGA	0.3036
X3PA	0.1694
X2P.	-1.9270
FT.	1.0195
TRB	0.4979
PF	-0.1388
PTS	0.2623
TS.	-5.9515
X3PAr	-2.5141
FTr	2.0851
TRB.	-0.0252
AST.	0.0454
STL.	0.1553
BLK.	0.1311
TOV.	-0.0197
USG.	0.1951
BPM	0.3175

Appendix E: Full coefficient summary for the lasso regression model

Table 18: Non-zero coefficient summary for the lasso regression model (13 coefficients). Salary measured in millions of U.S. dollars.

#	Variable	Coefficient
1	(Intercept)	-20.4670
2	PositionPG	0.4291
3	Age	0.7340
4	TeamCLE/MIA	6.0228
5	TeamLAC	3.2750
6	TeamLAL/LAC	14.6941
7	TeamUTA	-0.1579
8	X2P.	-0.0359
9	TRB	0.2939
10	PTS	0.9305
11	TS.	-0.0396
12	AST.	0.0179
13	BPM	0.0612